Riverbot Group Project

This project is a series of exercises that walk through different phases of an engineering project. Your group is to create an object-avoiding robot-car with the following features:

- Car must move
- Car must avoid obstacles

In addition, the customers prefer:

- Fast-moving cars
- Cars that don't break if they accidentally encounter obstacle
- Cars that travel in straight lines in absence of obstacles
- Extra features that differentiate the car

This team project constitutes **35%** of your course grade. Note that throughout the project, a portion of each grade will be based on the creativity of your solution. **"Replication" of the model will earn you no more than 50% of the project grade.**

This project comprises the 10 elements listed below. You will submit some items as a group, and you'll submit some items individually. Please note which are which below.

Item	Description	Weight	Date Due	Grading
A	Straight-travel Program Left-turn Program	5%	10/28/2026	Team
B	Obstacle-detection Program	10%	11/4/2026	Team
C	Preliminary Bracket Design	5%		
D	Requirement Specifications	10%		Individual*
E	Riverbot v0.1	10%	11/11/2026	Team
F	Concept Proposal	10%		
G	Demonstration	15%	11/18/2026	Team
H	Oral Presentation	15%	12/7/2026	Individual*
I	Technical Report	15%	12/9/2026	Team
J	Teamwork Survey	5%		Individual
Total: 100.00%				

* While these are done as a group, grading will be on an individual basis

Required Formats

- Each assignment should be submitted as a single PDF file
- All drawings should be done in CAD (except for drawings in the Concept Proposal)
- **Do not include a team member's name on the item if they do not contribute!**

Assignment Details

A. Straight Travel/Left Turn

Create programs that execute the following movements:

- Car must travel 2.0m without deviating by more than 0.1m
- Car must:
  - Travel straight for 1.0m ± 0.1m, and then
  - Stop and turn left by 90° ± 10°, and then
  - Travel straight for 1.0m ± 0.1m, and then
  - Stop for 5 seconds

B. Obstacle Detection

Create a program that executes the following movements:

- Car must travel straight travel towards an obstacle, then
- Detect the obstacle with the ultrasonic sensor and stop, then
- Turn right by 90° ± 10°, and then
- Travel straight for 1.0m ± 0.1m, and then
- Stop for 5 seconds

C. Preliminary Bracket Design

Use CAD software to design a bracket to mount your ultrasonic sensor to your car's servo motor. You may use digital calipers to measure your ultrasonic sensor.

D. Requirements Specification

- Write formal statements of 'what' your team's robot-car will do.
- Document must contain at least **12 requirements** ('*specs*').
- Annotate the formal requirement statements **with rationale** for each spec.
- Say how the requirement will be verified (inspection, analysis, test). Do not create specs that cannot be verified. For example, if you create a spec of "*Robot will be pretty*" or "*Robot will be easy to use*," you must specify how you plan to measure prettiness or ease-of-use.
- **Divide the requirements up among the team members, and each student turns in their specs individually (at least 3 specs per student). The preferred format is a table with 4 columns:**

No.	Spec	Rationale	Verification Method
1	Description (what) of spec #1	Rationale (why) for spec #1	Verification (how) of spec #1
2	Spec #2	Rationale for spec #2	Verification of spec #2
3	Etc		

E. Riverbot v0.1

- Demonstrate preliminary version of your robot-car. This will require:
  - Adding a servo-motor, ultrasonic sensor, and your custom bracket to your car
  - Program it to avoid obstacles

Demonstrate your car's ability to avoid obstacles via uploaded video (or in-class)

F. Concept Proposal

- Gather information, consistent with the Engineering Design Cycle that we reviewed.
- Brainstorm ideas; select and document ideas (select a scribe for your group).
- Draw sketches or diagrams of proposed solution with overall features.
- Include a PERT chart containing "Task Description", "Personnel", and any additional columns for each week of work on the project. **It must include at least 10 tasks.**
- **Submit a proposal consisting of 2-3 pages. The proposal should contain one or more diagrams (hand-drawn sketches are fine), along with paragraphs that describe how your proposed solution would work.**

G. Demonstration (Proof of Concept)

- You will showcase your car and demonstrate its abilities to the class.

H. Oral Presentation

- Your team presents your solution to the class.
- **All team members must speak** - each member receives an individual grade.
- **Duration of total presentation: 10 to 12 minutes**
- Presentation must include (in no particular order):
  - Review of requirements
  - CAD design of your custom bracket
  - Comparison of test results to requirements.
  - Comments on the features of your solution. These may include speed, object avoidance, ergonomics, safety, cost, ease-of-maintenance, etc.
  - Recommendations for future enhancements. In other words, if your team had more time, what further features or improvements would you add?
- Presentation will be graded on the following five criteria (20% each): professionalism, content, analysis, flow of presentation, and adherence to time limit.

I. Technical Report

- Hand in a written report that summarizes the design of the product. A typical length of such reports is 4-5 pages, excluding the title page and end notes.
- Sections in the report should include: Introduction, Product Requirements, Design Approach, Final Design, Test Results, Conclusions, and Recommendations for Future Development.

J. Teamwork survey

- This is a questionnaire on Blackboard for each team member to fill out individually.
- You will be asked how well the team functioned, how well it was organized, how well the tasks were distributed among members, and how well individual members performed.